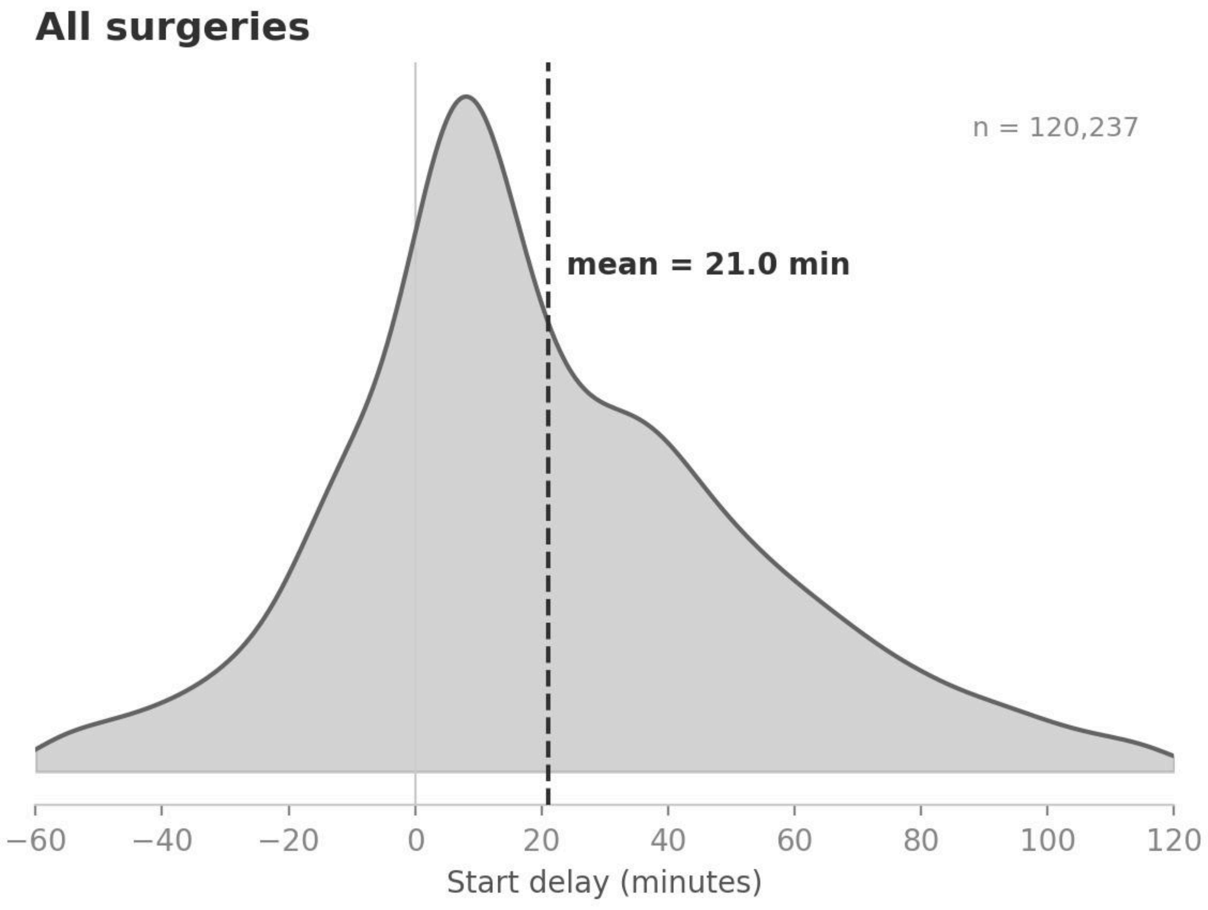
Why Surgeries Start Late: A Scheduling Fix for Rigshospitalet

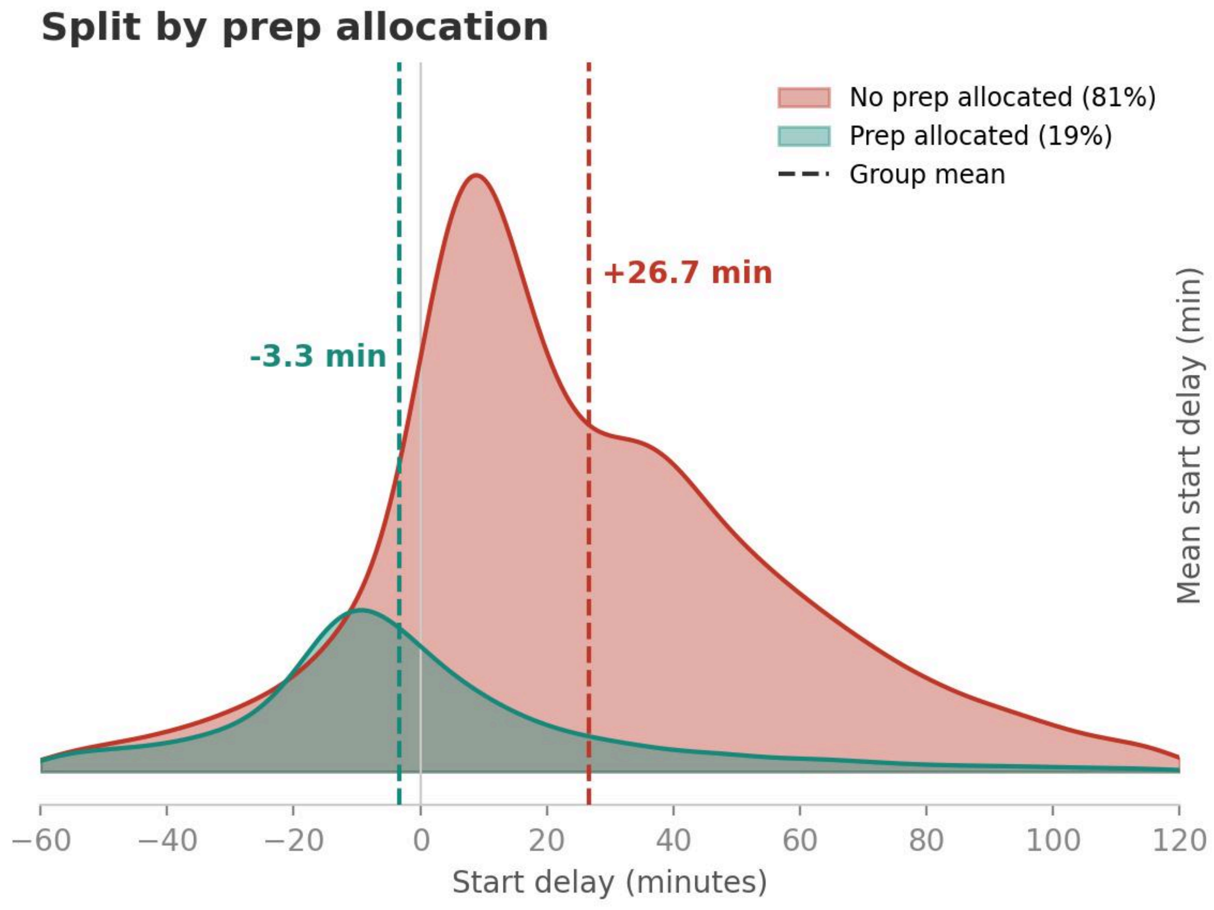
Authors: Linus Juni, Theodor la Cour, Mathilde Sørensen, Jose Diaz, Rune Isaksen & Alexander Sucur

Problem: The schedule allocates zero prep time for 81% of surgeries

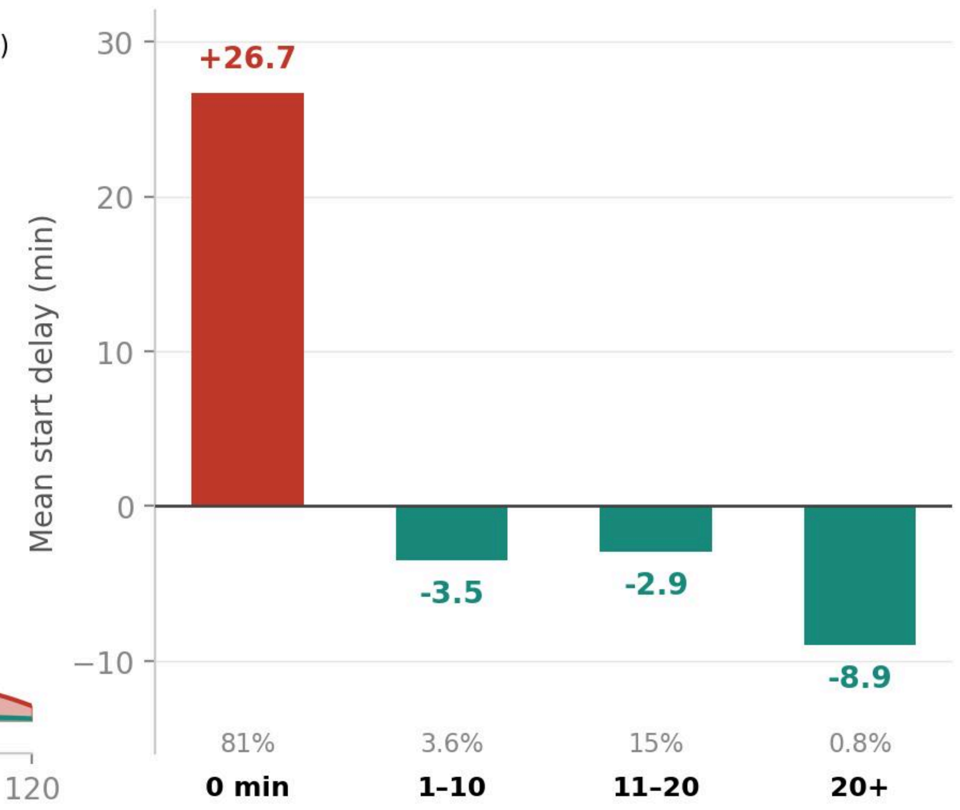
All surgeries


n = 120,237
mean = 21.0 min
−60 −40 −20 0 20 40 60 80 100 120
Start delay (minutes)

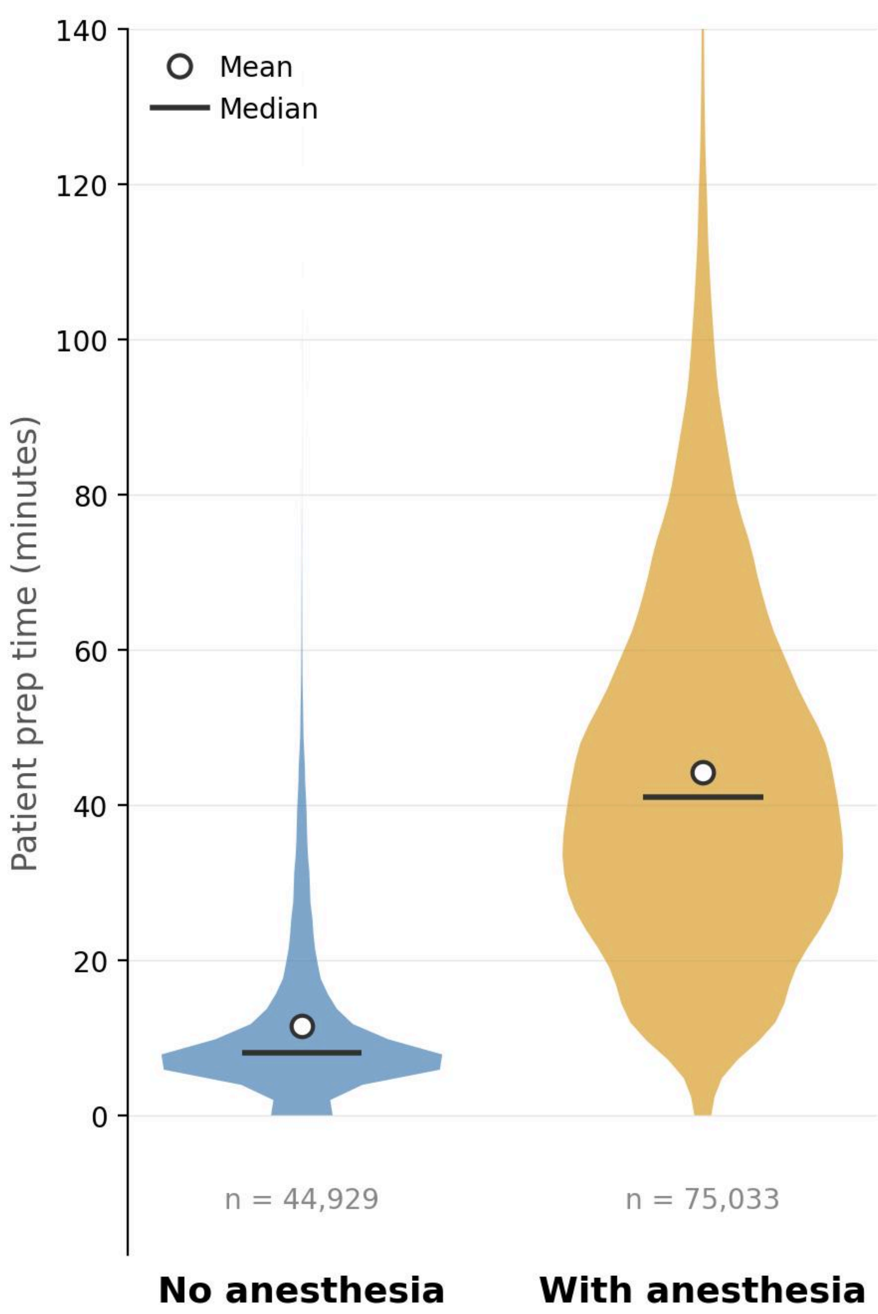
Split by prep allocation

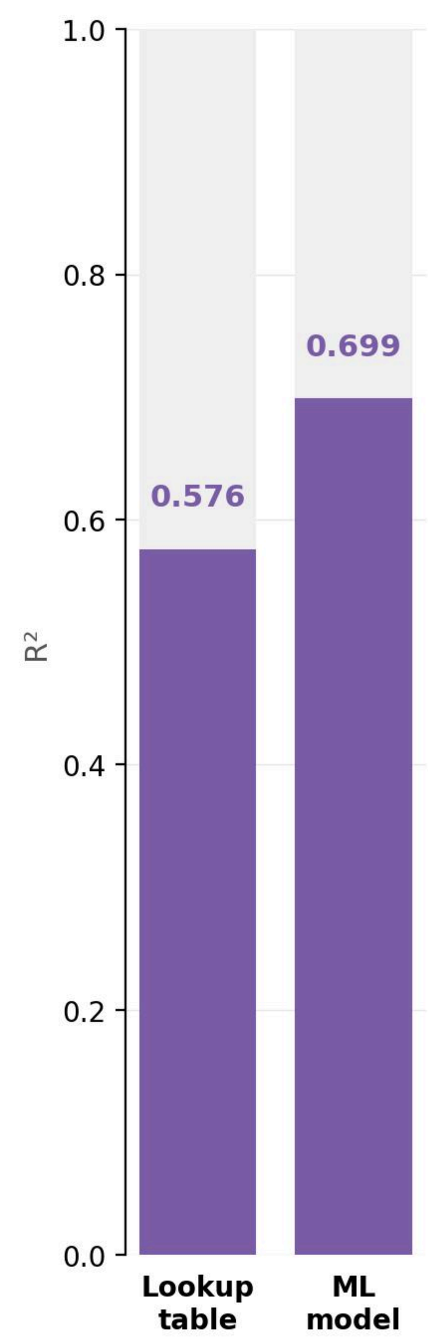

No prep allocated (81%)
Prep allocated (19%)
Group mean
-3.3 min
+26.7 min
−60 −40 −20 0 20 40 60 80 100 120
Start delay (minutes)

By prep bucket


Mean start delay (min)
30 20 10 0 −10
+26.7
-3.5
-2.9
-8.9
81% 3.6% 15% 0.8%
0 min 1–10 11–20 20+

Solution: A 64-cell lookup table predicts what the schedule ignores

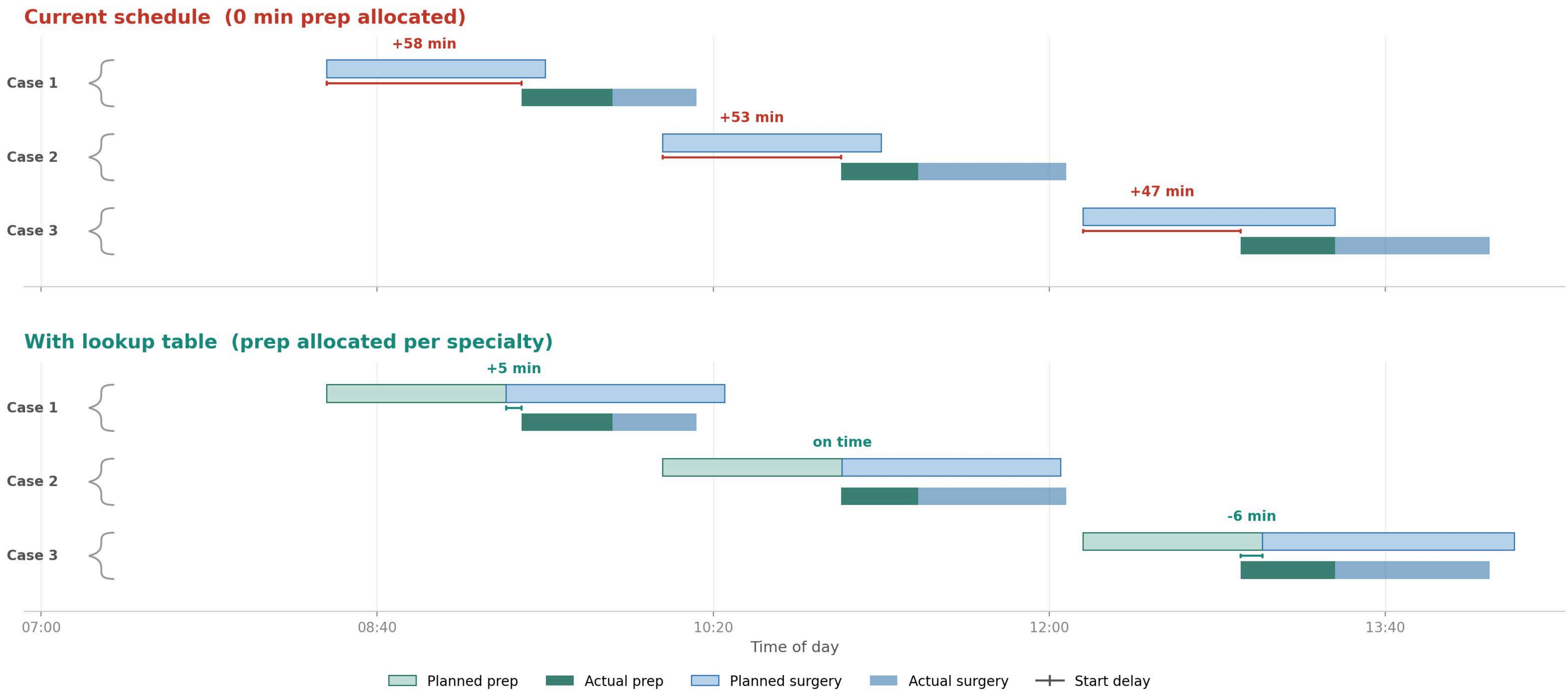

Mean
Median
Patient prep time (minutes)
140 120 100 80 60 40 20 0
n = 44,929
n = 75,033
No anesthesia
With anesthesia


R²
1.0 0.8 0.6 0.4 0.2 0.0
0.576
0.699
Lookup table
ML model

	No anesthesia	With anesthesia
Arthroplasty	12	80
Anesthesiology	32	40
Breast surgery	44	65
Pediatric surgery	34	33
Pediatric orthopedics	16	53
⋮ 22 more ⋮		
Trauma orthopedics	28	59
Tumor orthopedics	24	65
Urology	28	49
Ophthalmology	8	32
ENT	21	37

Mean prep time (min): 10 – 80

Impact: Same day, different schedule


Current schedule (0 min prep allocated)
Case 1
+58 min
Case 2
+53 min
Case 3
+47 min
With lookup table (prep allocated per specialty)
Case 1
+5 min
Case 2
on time
Case 3
-6 min
07:00 08:40 10:20 12:00 13:40
Time of day
Planned prep
Actual prep
Planned surgery
Actual surgery
Start delay

AI tools were used for coding assistance and ideation. All analysis, conclusions, and recommendations are the authors' own.